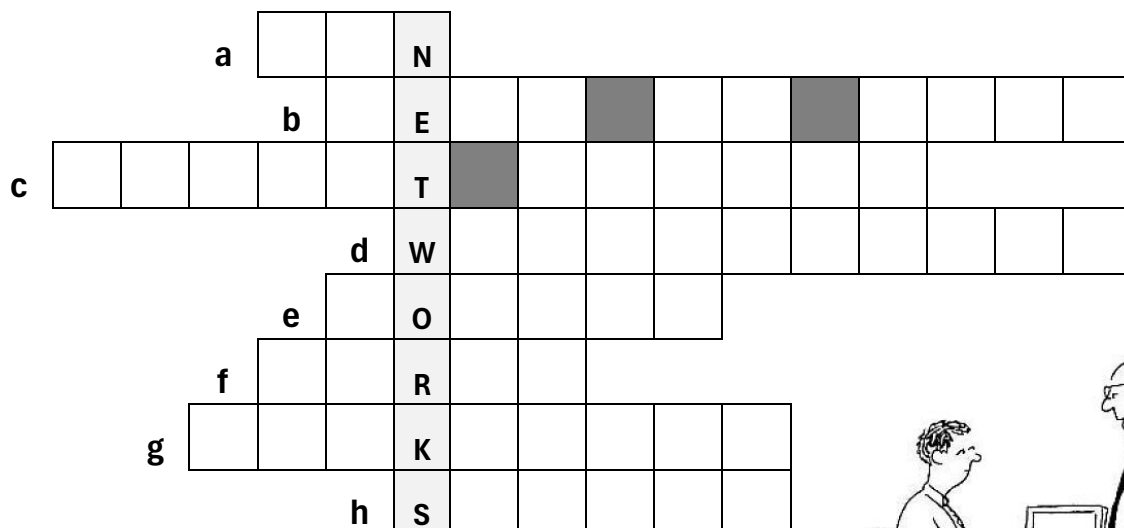


Unit 4 – Networks



1) Using the clues provided, solve the following crossword.

- Acronym for Local Area Networks,
- two computers have the same capabilities, or client-server,
- where one computer acts as the server containing the main hard disk and controlling the other workstations or nodes,
- a keyboard and screen with which a person can use a computer system, or an area in an office, factory, etc. where a single person works,
- a device that forwards data packets and is needed to link a LAN to another network (e.g. to the Net),
- (a piece of) thin metal thread with a layer of plastic around it, used for carrying electric current a place where things such as lines or systems join (*noun, plural*),
- high-speed lines in the main transmission paths within a WAN,
- a central computer from which other computers get information.



"When I was a student, wireless data transmission meant passing notes in class."

2) Read the following text carefully and answer the questions below (booklet, page 28).

Networks

LANs (Local Area Networks)

Networking allows two or more computer systems to exchange information and share resources and peripherals.

LANs are usually placed in the same building. They can be built with two main types of architecture: peer-to-peer, where the two computers have the same capabilities, or client-server, where one computer acts as the server containing the main hard disk and controlling the other workstations or nodes, all the devices linked in the network (e.g. printers, computers, etc.).

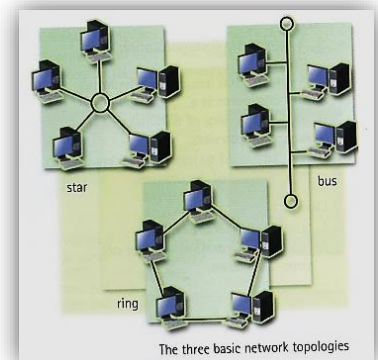
Computers in a LAN need to use the same protocol, or standard of communication. Ethernet is one of the most common protocols for LANs. A router, a device that forwards data packets, is needed to link a LAN to another network, e.g. to the Net.

Most networks are linked with cables or wires but new Wi-Fi, wireless fidelity, technologies allow the creation of WLANs, where cables or wires are replaced by radio waves. To build a WLA you need access points, radio-based receiver-transmitters that are connected to the wired LA, and wireless adapters installed in your computer to link it to the network. Hotspots are WLANs available for public use in places like airports and hotels, but sometimes the service is also available outdoors (e.g. university campuses, squares, etc.).

Network topology

Topology refers to the shape of a network. There are three basic physical topologies.

- **Star:** there is a central device to which all the workstations are directly connected. This central position can be occupied by a server, or a hub, a connection point of the elements of a network that redistributes the data.
- **Bus:** every workstation is connected to a main cable called a bus.
- **Ring:** the workstations are connected to one another in a closed loop configuration.



There are also mixed topologies like the tree, a group of stars connected to a central bus.

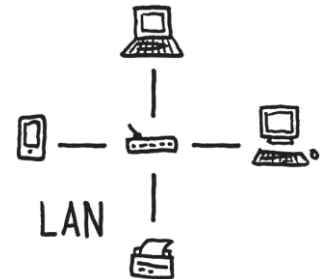
WANs (Wide Area Networks)

WANs have no geographical limit and may connect computers or LANs on opposite sides of the world. They are usually linked through telephone lines, fibre-optic cables or satellites. The main transmission paths within a WAN are high-speed lines called backbones.

Wireless WANs use mobile telephone networks. The largest WAN in existence is the Internet.

2.1. Correct the following statements (booklet, page 29).

- a) LANs link computers and other devices that are placed far apart.
- b) In a client-server architecture, all the workstations have the same capabilities.
- c) The word protocol refers to the shape of the network.
- d) Routers are used to link two computers.
- e) Access points don't need to be connected to a wired LAN.
- f) Wireless adapters are optional when you are using a WLAN.
- g) Hotspots can only be found inside a building.
- h) The Internet is an example of a LAN.
- i) Wireless WANs use fibre and cable as linking devices.



3. Use the words in the box to complete the sentences (booklet, page 29).

LAN | WLAN | nodes | peer-to-peer | hub | server | backbones

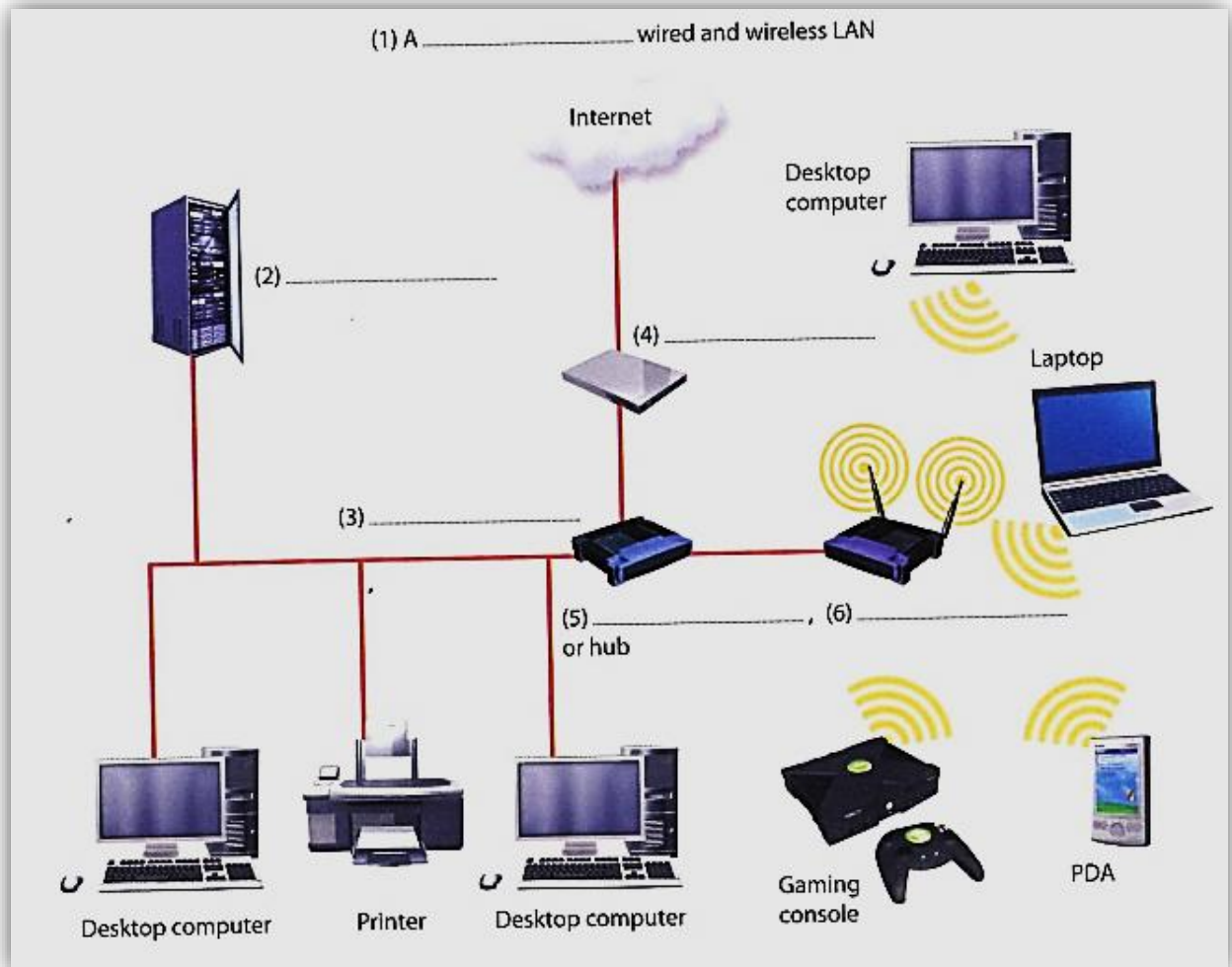
- a) All the PCs on a _____ are connected to one _____, which is a powerful PC with a large hard disk that can be shared by everyone.
- b) The style of _____ networking permits each user to share resources such as printers.
- c) The star is a topology for a computer network in which one computer occupies the central part and the remaining _____ are linked solely to it.
- d) At present Wi-Fi systems transmit data at much more than 100 times the rate of a dial-up modem, making it an ideal technology for linking computers to one another and to the Net in a _____.
- e) All of the fibre-optic _____ of the United States, Canada and Latin America cross Panama.
- f) A _____ joins multiple computers (or other network devices) together to form a single network segment, where all computers can communicate directly with each other.

4. Identify the topologies of communication networks described below (booklet, page 29).

- a) All the devices are connected to a central station. _____
- b) In this type of network there is a cable to which all the computers and peripherals are connected. _____
- c) Two or more star networks connected together; the central computers are connected to a main bus. _____
- d) All devices (computers, printers, etc.) are connected to one another forming a continuous loop. _____

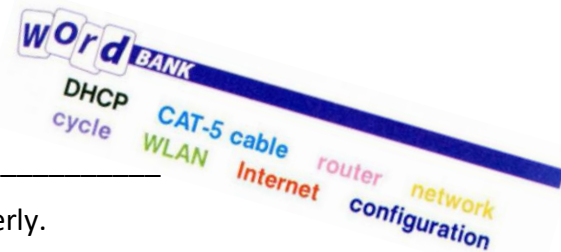
5) Label the elements of this LAN with the words given (booklet, page 26).

router – wireless AP (access point) – cables – mixed – central computer (file server) – broadband modem



6) Fill in the blanks with the correct words or phrases from the word bank.

- The _____ lets a network connect to the Internet.
- A LAN is a type of _____.
- Businesses communicate on the _____.
- We do not use cables for our network. We have a _____.
- Routers need a correct _____ to work properly.
- Businesses with a LAN use _____ to connect computers.
- It is sometimes necessary to _____ a network to make it work.
- The _____ provides different IP addresses for computers.





7) Choose the correct word in each of the following (booklet, page 26).

7.1. The speed with which a modem can process data is measured in _____.

- a) bandwidth b) bits per second (bps) c) signal

7.2. Cables consisting of several copper wires each with a shield are known as _____ cables.

- a) twisted pair b) optical fibre c) power cables

7.3. Computers that are connected together within one building form a _____.

- a) WAN b) ISP c) LAN

7.4. If you transfer a file from a remote computer to your computer, you _____.

- a) download b) upload c) run

7.5. To send out information is to _____.

- a) signal b) packet c) transmit

7.6. A document containing information and graphics that can be accessed on the internet is _____.

- a) a website b) a web page c) the World Wide Web

8) Here is a list of instructions for someone wanting to set up a small network. Put the instructions in the correct order.

- a) Make wiring and layout plans for your network. _____
- b) Hook up the network cables by connecting everything to the hub. _____
- c) Check that each computer has an IP address and give it a name. _____
- d) If you're installing a small network, twisted pair will be adequate. However, in order to span greater distances and to minimize magnetic and electrical interference use fibre optic cable. _____
- e) Decide on the type of network you want to install. To enable you to transfer large amounts of data, choose Fast Ethernet (100BaseT). _____
- f) Install network adapters in the computers. _____
- g) Add an internet gateway to your network to set up a shared internet connection. _____
- h) Install driver software for the adapter driver and install client software to share printers and files. _____
- i) Check which protocols are installed and add any other protocols you require. _____
- j) Get the hardware you need: an Ethernet adapter card for each computer that doesn't have an Ethernet port, a hub if you've got more than two computers, cables and wall jacks. _____

For further practice, solve exercises from page 25 in your Booklet.



What Are Prefixes?

A prefix is a group of letters that is added to the beginning of a word (nouns, adjectives, verbs, etc.) to change their meaning and use.

Prefix	Meaning	Examples
Inter -	between	<i>interactive, international</i>
Down -	reduce or lower	<i>downgrade, downhearted</i>
Trans -	across	<i>transatlantic, transfer</i>
Up -	make or move higher	<i>upgrade, uphill</i>
Intra -	inside or within a place or group	<i>intra-European</i>
Com Con	together, with	<i>combination, community</i> <i>conspiracy, consortium</i>

Let's practise!

Complete the words in the following sentences by adding the prefix *inter-*, *intra-*, *trans-*, *com-*, *con-*, *up-* or *down-*.

- Last month computer _____ time cost the company over €10,000 in lost production.
- The computers in the production department have now been successfully _____ connected with those in the planning department.
- Once you have completed payment details the data will be _____ mitted via a secure link.
- We cannot network these computers because the systems are not _____ patible.
- Many companies distribute internal documents on their own _____ net.
- Once the home page has been completed, we'll be ready to _____ load the site.
- Cables are being laid throughout the building as the network requires physical _____ nections.
- Using the network he was able to _____ bine the data from different reports.